Proceedings of the 14th IFAC Symposium on
Information Control Problems in Manufacturing
Bucharest, Romania, May 23-25, 2012

Reducing intermodal transportation impacts on society and environment by path selection: a multiobjective shortest path approach

Marie SAWADOGO[*], Didier ANCIAUX[**], Daniel ROY[***]

**, ** LGIPM, University Paul Verlaine of Metz, France (e-mail sawadogo, anciaux@univ-metz.fr)*
**** LGIPM, ENIM, France (e-mail roy@enim.fr)*

Abstract: There is no doubt that the transport sector contributes significantly to the overall energy consumption and emission of greenhouse gases. The transportation systems are increasingly related to environmental problems. The present contribution aims to take into account the direct and indirect impacts of transportation systems whatsoever on the environment or on the society. This paper proposes the reduction of environmental and societal impacts of intermodal freight by using multiobjective shortest path approach. A novel algorithm called Multiobjective Ant colony Algorithm for Multiobjective Shortest Path Problem (MOSPACO) is presented. This algorithm aims to help decision makers to choose a path with less environmental impacts in an intermodal network. The ant colony optimization metaheuristic is used to choose the 'best' path in the network, which means the path with the best compromise economy/environment. A numerical analysis is then presented based on a network build using real geographical data and the obtained non-dominated solutions are analyzed.

Keywords: Intermodal Transportation, Environmental Impacts, Green supply chain, Multiobjective Optimization, Multiobjective shortest path

1. INTRODUCTION

Transportation has become an essential component of almost any modern activity. In a century and a half, freight transports have been multiplied by 1000 to the globe. Moreover, the increases of distances and flows between firms have created an increased demand for intermodal transportation services and multimodal transport infrastructures. The transport sector is the largest consumer of petroleum products and the largest emitter of greenhouse gas. Thus, national and international sustainable development strategies highlight the need to reduce the level of energy consumed by transportation and to move towards transportation modes with less environmental impacts such as railways and waterways.

Besides the challenges of energy and climate change, the supply chain is also entering an unprecedented age of opportunity. In particular, the modern supply chain still enjoys relatively inexpensive and abundant energy inputs and raw material resources, and is also more connected than ever before (Beamon, 2008).

Many factors are then involved in decision making in a transportation system. Traditionally, studies on transportation are based on finding a path in a graph considering a single objective; travel cost or travel time most of the time. However, in most real world problems the decision making is based on multiple criteria. In our case, we aim at finding a path with less environmental and societal impacts while minimizing transportation cost and time. The environmental criterion is no longer a constraint to the problem, but a goal to be achieved. Hence the need of using multiobjective optimization methods to find the path with the best compromise cost/impact. Some companies are asking if they really need to make these changes; as demonstrated in (Sawadogo and Anciaux, 2009) making a right choice of transportation mode have many benefits in terms of impacts reducing and economic growth. It is proposed in (Quariguisa et al., 2009) a method to find a trade-off between the criteria preferred solutions for business and the environment, in order to answer to the question 'how much do we have to spend in order to improve environmental quality?', assesses the risks and benefits of greening a company's supply chain.

Taking these concerns into account, the present study advances the use of ant colony metaheuristic for multiobjective shortest path problem solving, by first evaluating transportation the carbon footprint in a supply chain and the other impacts while providing a limit degradation of the economic benefits. Our contribution for this study includes a literature review in section 2, a presentation of MOSPACO algorithms in section 3, an application to a case of intermodal transportation in section 4 and section 5 concludes the paper and suggests some future research directions.

2. PRELIMINARIES

2.1 *Purpose and scope*

Intermodal freight is the subject of increased interest because of problems including traffic congestion, environmental concerns and safety issues on the routes. An awareness of the strategic importance of greening the supply chain constrain companies to reconsider the traditional organization of the supply chain with the intention of doing a supply chain able to combine economic issues and environmental issues. Thus,

978-3-902661-98-2/12/$20.00 © 2012 IFAC

10.3182/20120523-3-RO-2023.00063

research is increasingly undertaken for environmental assessment of logistic activities and for implementing methods for impacts reduction. Therefore in an intermodal network with several alternative paths and transportation modes, choosing path to convey goods from an origin point s to a destination point t must simultaneously optimize the transportation time, internal and external costs, performance and also the environmental and societal impacts. Hence we need to use multiobjective optimization methods to find the path with the best compromise solution. The final result must be a path with less environmental and societal impacts while minimizing transportation cost and time. The transportation modes concerned by our study are rail, air, road, inland water and maritime mode.

2.2 *Green supply chain and transportation impacts*

The current state and trend of environmental degradation (from regulatory, consumer, and moral standpoints) indicate a need for a change in manufacturing and supply chain philosophy. That is, there must be a fundamental shift in the way production systems operate. There must be a move towards sustainability, achieved through waste reductions in resource use and waste generation, and a move away from one-time use and product disposal (Beamon, 1999). Green supply chain recognizes the impacts of logistic activities on the environment and the society, and tries to find ways to reduce them. The definition and scope of green supply chain management in the literature has ranged from green purchasing to integrated green supply chains flowing from supplier to manufacturer to customer, and even reverse logistic (Srivastava, 2007). The green supply chain management can be define as 'integrating environmental issues into supply-chain management, including product design, material sourcing and selection, manufacturing processes, delivery of the final product to the consumers as well as end-of-life management of the product after its useful life'. According to (Klassen et al., 2004), five method exist for green supply chain management; environmental certification, environmental pollution prevention, reverse logistics, life cycle analysis and green-design.

Transportation activities are dominant factors of greenhouse gas emissions and thus their impacts on the environment must be analyzed and actions must be taken to mitigate them. These impacts, like all environmental impacts, can fall within three categories: direct impacts which are the immediate consequence of transport activities on the environment, indirect impacts which are the effects of transport activities on environmental systems and cumulative impacts which are the additive, multiplicative or synergetic consequences of transport activities.

Intermodalism is a transportation process through interconnection of networks, involving various combinations of modes in which all components are perfectly linked to form an effective coordination of flows. This organization offers a full range of transport modes and multiple transportation options, which can coordinate the deliveries, the manufacturing, stockpiling and the different distribution functions. The intermodal networks are characterized by dynamic changes and multiple transportation modes, operating simultaneously between them (Qu et al., 2008). Intermodal transportation systems offer a wide choice of transportation mode and several alternative paths, hence the need for better coordination of flows and movements in such a system.

The impacts of transport on health and the environment are well known today; they includes global warming, the deterioration of the ozone layer, the dispersion of organic and inorganic substances including toxic ground-level ozone, the scarcity of oil and other natural resources, and the degradation of landscapes and soils. Furthermore, the transport sector is the most widespread source of noise in all countries and the most prevalent cause of annoyance and interference. Therefore, traffic noise reduction measures have the highest priority. The noise generated by transportation systems affects people in residential areas; these nuisances are important since the traffic is important. Conditions other than distance (e.g. topography, trees, and buildings) can also influence noise levels (Forkenbrock, 1999). Concerning the accident risk, freight vehicle accidents are life threatening in many areas. So there have been innumerable studies on freight vehicle risk analysis (Chapman et al., 1998).

Table 1 summarizes the major impacts of intermodal transportation systems.

Table1: Major environmental impacts per transportation mode

Transportation mode	Environmental and societal impacts
Air transportation	Noise, engine emissions, waste disposal, CO_2 and water vapor for kerosene combustion, hydrocarbons, CO and nitrogen oxide (NOx) from incomplete combustion, contaminate soil, groundwater, and surface water from vehicle maintenance and refurbishing.
Rail transportation	Ground water and soil contamination from solvents, acid waste, vibrations caused by passing trains, air pollution from fuel vapors, solid waste disposal problems from rail terminal operations, fines particulates from engines, Noise, accidents.
Truck transportation	Air pollution from fine particulates, hydrocarbons, carbon monoxide and fuel emissions, NOx, Sulfur components. Water contamination from sludge. Waste disposal from terminal operations. Noise, accidents, traffic congestion.

Maritime transportation	Water contamination from blasting media and heavy metal. Water contamination from turbine oil, lubricants, degreasers. Air pollution from particulate matter in cargo handling. Water and soil contamination from fuel spills.
Multimodal hubs	Wetland displacement, green space displacement, lost habitat, land occupancy, congestion.

Bringing environmental issues into a transportation system changes the configuration of a supply chain and brings a need of changes in logistics infrastructures. Therefore, we must find sources for improving desired environmental performances without sacrificing the transportation cost and time. In recent years, several studies have been published for impacts abatement in transportation systems. Most of these studies are partly on the analysis of impacts from transportation systems and the literature review on methods to reduce impacts and partly on the assessment of environmental costs of transport as presented in (Forkenbrock et al., 1999). Some models for computing the environmental and social impacts of transportation systems can be viewed in (Janic, 2007). Piecyk and McKinnon (Piecyk and McKinnon, 2010) studied three scenarios illustrating this trend for road freight. According to their study, the carbon footprint of the road freight sector will increase to 30.0 million tonnes of CO_2 in 2020 (56% above the present level) if no action is implemented to limit the phenomenon.

Regarding contributions to impact abatement, the introduction of new technologies and new transport concepts such as alternative fuel, electric vehicles, eco-driving, early morning-distribution system, vehicle utilization and avoidance of empty trips, environmental Management Systems (ISO 14000, ISO 14001) may offer reduced emissions per vehicle, or encourage switching to more environment-friendly modes. Another way of reducing environmental impacts is to persuade travellers to minimize the number of vehicle trips by switching to non-motorized modes. In (Rondinelli et al., 2000), they proposed an integrated 'proactive environment management system' which seeks to prevent pollution and eliminate sources of environmental degradation of intermodal freight. In addition to all these methods we must add that an alternative to reduce environmental and social impacts is to choose a judicious path and transportation mode for shipments. Thus, our contributions for the impact of intermodal transportation impacts reducing is through the proposal and design a system for decision support for choosing a path minimizing impacts.

2.3 *Multiobjective optimization and multiobjective shortest path problem*

Multiobjective optimization, also called multicriteria or multi-attribute optimization is a process of optimizing simultaneously two or more conflicting objectives, subject to some constraints. Indeed, most of the decisions in real life are based on several criteria. Multiobjective optimization problems are applied in various areas including product design / process, manufacturing, automotive, finances ... the optimal solution in this case is obtained by a compromise between the objectives.

2.3.1 *Multiobjective shortest path problem*

The shortest path problem is one of the most studied network optimization problem since the fifties (Dijkstra, 1959). These studies are based on the finding a path in a graph based on a single objective; travel cost or travel time most of the time.

However, in most real world problems, the complexity of the social and economic environment requires to take into account several criteria other than the cost and the time. It follows the development of methods most likely to take into account simultaneously several criteria, hence the concept of multiobjective shortest path problem.

If G is the network considered, let n be the number of criteria, with $n \geq 2$. At each arc $(i,j) \in A$ is associated a vector function c of dimension n, which assigns to each arc a 'cost' c_{ij}^{l}, $(l = 1, \dots, n)$ representing all the costs on the arc (i,j) for the lth criteria. Where,

$$c: \begin{array}{l} A \longrightarrow \mathbb{R}^n \\ (i,j) \mapsto c^l(i,j) = c_{ij}^{l} = \left(c_{ij}^{1}, \dots, c_{ij}^{n}\right) \end{array} \quad (1)$$

The multiobjective shortest path problem is defined as:

$$min\, f^l(x) = \sum_{(i,j)\in A} c_{ij}^{l} \,.\, x_{ij} \qquad \forall l = 1, \dots, n \quad (2)$$

Many articles have been written about this problem and most of them concerned problems based on two criteria. Indeed, some dynamic programming approaches have been applied for this kind of problem (Kostreva et al., 1993). The resolution methods in the literature are essentially label setting methods (Martins, 1994; Caramia et al., 2009; Gandibleux et al., 2006), label correcting methods (Skriver et al., 2000, Guerriero et al., 2001), k-shortest paths algorithms (Androutsopoulos et al., 2008; Martins, et al., 1998), utility functions methods (Modesti et al., 1998).

Evolutionary algorithms are increasingly used to solve multiobjective shortest path problem. The major evolutionary algorithms used for solving multiobjective shortest path problems are ant colony algorithms and genetic algorithms. In (Pangilinan et al., 2007) SPEA-II is applied to a randomly generated network with nine possible configurations; they obtained a Pareto front with the non-dominated paths in a polynomial time. Recently, some researchers have designed Ant Colony Optimization (ACO) algorithms to deal with multi-objective shortest path problems; in (Ghoseiri et al., 2010) it is demonstrate that ant colony optimization algorithms are better than label correcting algorithms for multiobjective shortest path problem solving; however, most of the existing methods aims at solving bi-objective problems (here we have seven conflicting objectives) and don't take into account the intermodal network, when intermodality increases the problem complexity.

2.3.2 *Ant colony optimization*

One of the most recent evolutionary algorithms is the ant colony optimization (ACO) paradigm that was first introduced by Marco Dorigo (Dorigo et al., 1991). The ant

colony algorithm algorithms are inspired by ant behavior. Artificial ants in ACO implement a random heuristic construction which builds probabilistic decisions based on the artificial pheromone and the input data to solve the problem; for details on this method see (Dorigo et al., 2005). Given the success of ACO algorithms in single objective areas, many researchers have been interested to use them for solving multi-objective problems. The excellent speed of ACO in solving combinatorial optimization problems has been proven; in (Garcia-Martinez et al., 2007) it is demonstrated that multiobjective ant colony algorithms are more efficient than existing genetic algorithms for multiobjective problems, so there are now employed as optimization tools instead of exact methods for solving multi-objective problems. Many studies are increasingly conducted on applying the ant colony optimization for multiobjective problems. They designed various kinds of algorithms by modifying the procedures of original ant colony algorithms and other related algorithms. Most of them are specific proposals to solve a concrete multi-criteria problem such as scheduling (Gravel et al., 2002), vehicle routing problem Baran et al., 2003), or portfolio selection Doerner et al., 2003), among others; some of them propose algorithms based on several colonies, others on a single colony. These existing algorithms differ from each other on five main points; *Number of colony* (multiple colony or single colony), *Pheromone strategy* (Multiple or single matrix), *Heuristic information strategy* (Multiple or single matrix), *Solution construction and evaluation* (targeted, dynamic or fixed), *Pheromone update (partial or total)*, see (Alaya et al., 2007; Angus, et al., 2009; Mariano et al., 1999; Gambardelta et al., 1999) for more details on these algorithms.

3. OUR APPROACH: DESCRIPTION OF MOSPACO ALGORITHM: MULTIOBJECTIVE ANT COLONY OPTIMIZATION FOR MULTIOBJECTIVE SHORTEST PATH

3.1 Problem description

In an intermodal graph G, we consider the transportation of a quantity Q tonnes of good from an origin node s to a destination node t. We have in this graph a wide choice of paths and transportation mode as presented in figure 1. The goal here is to determine the 'most efficient' path from s to t in order to minimize the transportation cost, the travel time, the environmental and the societal impacts. The criteria for the decision making are so numerous and are divided into three groups:

- *The economic or industrial criteria*: the 'transportation cost', the 'travel time' and the 'the damage due to transhipments'. The travel time takes into account the transhipment time including the unloading/loading time, the average travel time between two nodes in the network and the delays due to congestion. The transportation cost includes the shipment cost (fixed and variable costs), the filling penalty, the transhipment cost, the in-transit inventory cost, the discount granted by the carrier based on the number of means of transport, and the time penalties. The damages due to transhipment represents the amount of damage occurring during a transhipment of each tonne of product at node i of the path $\mathcal{P}_{st}$
- *The environmental criteria*: the 'air pollution' and the 'energy consumption'. The air pollution represents the assessment of the amount of CO_2, NO_x and VOCs, the main gas emitted by transportation modes. The total energy consumption is computed for each transportation mode taking into account the energy sourcing, production and the energy necessary for each transportation mode haulage.
- *The societal criteria*: the 'noise' and the 'risk of accidents'. We defined the accident risk as the probability for a conveyance that carried a quantity Q of goods over a distance d_{ij} to be involved in an accident on an arc (i,j). Noise level depends on the distance between the traffic lane (road, railways...) and the first homes or first impact site.

We consider here a problem of determining paths that minimize all these criteria, each of the objective function is modelled as presented in (Sawadogo and Anciaux, 2010).

3.2 Network

Let us consider a network as: $G = (V, A, M)$ where;

- $V = \{1, \dots, N\}$ is the finite set of nodes in the graph
- $A = \{(i_1, j_1), \dots, (i_L, j_L)\}$ represents the finite set of edge connecting nodes of G
- $M = \{m\} = \{r_{oute}, e_{au}, a_{ir}, r_{ail}\}$ the finite set of transportation mode available on the network

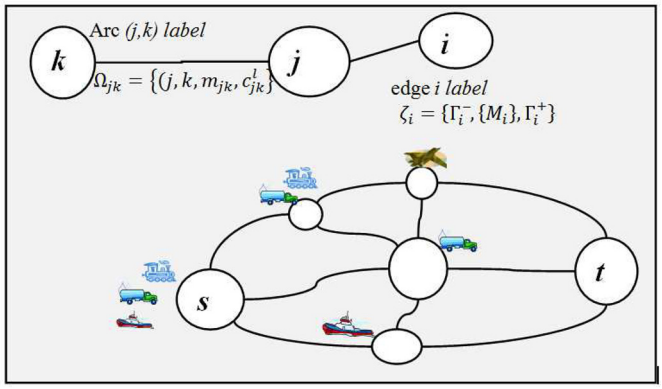


Fig. 1 The studied network

Let n be the number of criteria, with $n \geq 2$. At each arc $(i,j) \in A$ we associate a vector function c of dimension n, which assigns to each arc a 'cost' $c_{ij}^l, (l = 1, \dots, n)$.Where

$$c: \begin{array}{l} A \to \mathbb{R}^n \\ (i,j) \mapsto c^l(i,j) = c_{ij}^l = \left(c_{ij}^1, \dots, c_{ij}^n\right) \end{array} \quad (3)$$

The multiobjective shortest path problem is known as NP-hard and algorithms encountered in the literature are facing

particular difficulties for managing the large number of data about the paths and that results in a considerable computation time. The concept of 'optimal' solution becomes less relevant for optimization with multiple objective functions. Therefore, the resolution of such a model is to find a set of solutions called Pareto-optimal set or efficient set, using relationships called dominance relationships. Finding a path in the intermodal graph $G(V, A, M)$ is therefore to find the combination of node and transport mode to use. A path P_{st} between s and t is a sequence of nodes and modes: $P_{st} = \{(i_1 \equiv s, m_{i_1}, i_2), (i_2, m_{i_2}, i_3), \dots, (i_\kappa, m_{i_\kappa}, i_\kappa \equiv t)\}$ where $\kappa \in N$ is the number of nodes used on the path P_{st}. According to the combination between the edge parameters, each path is evaluated by a 'performance vector' or 'path cost vector' $c^l(P_{st}) = \{c^1(P_{st}), \dots, c^n(P_{st})\}$ which corresponds to the sum of costs associated with each criterion l for a path P_{st}. Hence,

$$c^l(P_{st}) = \sum_{(i,j)\in P_{st}} c_{ij}^l \quad (4)$$

Solving the problem defined here is therefore to find all paths with minimum value of the objective vector $c^l(P_{st})$.

3.3 MOSPACO algorithm description

In the MOSPACO (Ant Colony Optimization for Multi-Objective Shortest Path) algorithm, we proposed a new definition of the heuristic information (visibility) and the pheromones trails. Indeed, a matrix is associated with each objective and is defined as the inverse of the objective value for the arc to choose from. In other words, we used multimodal multiobjective pheromone trail and heuristic information, ant using them for modal switching and modal choice. Each ant counts the number of transshipment, when the selected transportation mode is different from the precedent. Each ant has a memory $Liste_f$ which contains information about the chosen path, the cities, the transportation mode used to go from a city to another city, and the objective vector containing the value of each objective computed for the chosen path. These vectors are then evaluated using non-domination sorting. The main contribution of this algorithm lies in changing the strategy of seeking a solution, the definition of visibility, and pheromone trail definition. Generally any ant algorithm must contain the following elements: heuristic information (visibility), pheromone updating rule, construction of solutions, selection probability, and termination condition.

3.3.1 Compute MOSPACO parameters

Heuristic information: This information indicates the desirability of moving from node $\boldsymbol{i}$ to node $\boldsymbol{j}$ and is calculated using a heuristic approach. There are various methods to evaluate the desirability of each arc. The heuristic information is calculated for all moves in all ants, it may significantly reduce the efficiency of the algorithm, thus it should be computed efficiently. For the implementation of ant algorithms, the heuristic information is define either 'a priori' i.e. the heuristic information is computed first at the beginning and remains unchanged during the running of algorithm or 'a posteriori' i.e. the heuristic information is dynamic and depends upon the current state of the ant. Our approach tries to combine these two aspects; the heuristic information is define at the beginning of the algorithm and update dynamically gradually as we advance in the search for solutions. Seven heuristic parameters $\boldsymbol{\eta_{ij}^{lm}}$ are used to solve the problem, one for each objective $\boldsymbol{l}$. These heuristic parameters are defined as the inverse of the criteria $\boldsymbol{l}$ for each transportation mode $\boldsymbol{m}$ used:

$$\boldsymbol{\eta_{ij}^{lm}} = \left(\frac{1}{\varepsilon + c_{ij}^l}\right) \quad (5)$$

In this formula $\boldsymbol{\varepsilon}$ is a small positive number added for avoiding division by 0. Under these conditions, for example the matrix associated to the travel time helps ants to favor arcs with less travel time and this is the same for the other criteria. In other words, if the value of the criteria associated to an arc is small, the attractiveness for this arc will be high and this move will be preferred to the others.

Pheromone trail: Seven pheromone matrixes are defined. When the process starts, the pheromone trails are set equal for each arc, with the initial value $\boldsymbol{\tau_0} = \frac{1}{N * n_tours}$ where $\boldsymbol{N}$ is the number of city and $\boldsymbol{n_tours}$ the maximum number of tours for all the ants in the colony. The pheromone trails are updated both locally and globally. The quantity of pheromone leaves by an ant that went through $(\boldsymbol{i}, \boldsymbol{j})$ is $\Delta\boldsymbol{\tau_{ij}^{lm}}$; this quantity is calculated as the inverse of value of the criteria $\boldsymbol{l}$ associated to the path determine by the ant $\boldsymbol{f}$ (6). This value is defined according to the transportation mode $\boldsymbol{m}$ used on $(\boldsymbol{i}, \boldsymbol{j})$, we then have a three dimensional pheromone matrix τ[origin node, destination node, transportation m],

$$\Delta\boldsymbol{\tau_{ij}^{lm}} = \frac{\boldsymbol{Q}}{c^l\left(\varepsilon + P_{st}^{(f)}\right)} \quad (6)$$

Where $\boldsymbol{Q}$ is a constant related to the amount of pheromone laid by ants $\boldsymbol{Q} = \boldsymbol{100}$, and $\boldsymbol{\mathcal{W}}$ the set of ants in the colony.

3.3.2 Solution procedures

Path construction: Let $\boldsymbol{\Gamma_i^+}$ be the set of outgoing nodes from node $\boldsymbol{i}$ that have not been previously visited by the ant $\boldsymbol{f}$. $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are parameters with values between [0,1] that represents the relative importance between (resp.) pheromone trail and heuristic information. All ants are placed on the source node $\boldsymbol{s}$ and their memory $\boldsymbol{liste_f}$ takes as initial value the node $\boldsymbol{s}$. When an ant $\boldsymbol{f}$ is on a node $\boldsymbol{i}$, it chooses the next node $\boldsymbol{j} \in \boldsymbol{\Gamma_i^+} / \boldsymbol{j} \notin \boldsymbol{liste_f}$ and the transportation mode $\boldsymbol{m}$ according to a probabilistic transition rule $\boldsymbol{p_{ij}^{fm}}$ (7). The node $\boldsymbol{j}$ with the highest probability $\boldsymbol{p_{ij}^{f}}$ is chose and added to $\boldsymbol{liste_f}$. This process is repeated for all the ants in the colony. An ant $\boldsymbol{f}$ stops the search of solution when it reaches the destination node $\boldsymbol{t}$ and is extracted from the colony, the others ants continue the search. The algorithm is repeated until the termination condition is met; the stopping condition here is a maximum running time and/or the total number of tours that must be fixed in advance.

$$p_{ij}^{fm} = \begin{cases} \dfrac{\left[\prod_{l=1}^{n}\left(\tau_{ij}^{lm}\right)^{\varepsilon_l}\right]^{\alpha}\cdot\left[\prod_{l=1}^{n}\left(\eta_{ij}^{l}\right)^{\pi_l}\right]^{\beta}}{\sum_{k\in\Gamma_i^+}\left[\prod_{l=1}^{n}\left(\tau_{ik}^{lm}\right)^{\varepsilon_l}\right]^{\alpha}\cdot\left[\prod_{l=1}^{n}\left(\eta_{ik}^{l}\right)^{\pi_l}\right]^{\beta}} & if\ j,k \in \Gamma_i^+ \\ 0 & otherwise \end{cases} \quad (7)$$

Γ_i^+ is the set of outgoing nodes from node i that have not previously visited by the ant f; the parameters ε_l and π_l are the relative importance of criteria against each other defined using an AHP hierarchy (Sawadogo and Anciaux, 2009) by pairwise comparison of criteria based on the scale of Saaty (Saaty, 1990). These parameters are the weight w_l associated to criterion l such as $\sum_{l=1}^{7} w^l = 1$, and defined according to two scenarios:

- *Industrial scenario*: the criteria 'transportation cost', 'travel time', 'damage due to transshipment' are the most important, then their weights are high than the other criteria and the heuristic information associated to them is most important.
- *Ecological scenario:* the criteria 'air pollution', 'energy consumption', 'accident risk', 'noise pollution' are the most important, then their weights are high than the other criteria and the heuristic information associated to them are most important.

Local update: the local updates are performed during the paths construction step. It is applied to the arc which is selected; thus, the local update rule reinforces the pheromone trails of the selected route, when the ant moves. It is defined as follows:

$$\tau_{ij} \leftarrow (1-\chi)\tau_{ij} + \chi.\tau_0 \quad (8)$$

τ_0 is the initial value of pheromone trail, χ the local update evaporation rate.

Global update: the global update is operated at the end of each iteration, after N_f ants have constructed solutions. The pheromone matrix is updated by the iteration best ant. The global update rule adds pheromone to the iteration best path. This best path usually indicates the shortest path, in our case the path with less environmental impact for ecological scenario or the path with less cost and time for industrial scenario. This update is performed by the following equation:

$$\tau_{ij}^{lm} = \rho\tau_{ij}^{lm} + \Delta\tau_{ij}^{lm}, = 1\ldots n \quad (9)$$

ρ is the global update evaporation rate.

3.3.3 The algorithm

Initialization
1. Set values of parameter α, β, ρ, τ_0, N_f *(number of ants) and termination condition then load network,*
2. Initialize the quantity of pheromone of all edges, iteration=1
3. for $f=1,\ldots,N_f$
Place all ants in the colony s
$liste_f \leftarrow s$
Construction
4. For *each* $j \in \Gamma_i^+$
For each transportation mode m *and for each criterion* $l = 1\ldots n$
Compute p_{ij}^{fm} *for each successor* j; $j \in \Gamma_i^+$
Choose the next node j *with the highest value of* p_{ij}^{fm} *(Eq. 7)*
Local update
End for
$tabou_f \leftarrow j$
Evaluation
5. *If ant arrive at the terminal node* t *then*
$P_{st} \leftarrow tabou_f$
Compute objective functions corresponding to the constructed solution.
Apply non-domination test to the solutions, update Pareto front
Pheromone updating
Update the pheromone quantity. (Eqs. (8) and (9))
If termination condition is met, stop otherwise got to step 4 and iteration ++.

3.3.4 Non-dominated sorting

Let N_D denote the set of non-dominated solutions and $S_{s,t}$ the set of solution for the current iteration. Solutions are evaluated based on the two scenarios defined above, for each scenario; the objectives are compared in pairs. In each of the iterations, each ant of the colony generates a solution. The current solutions obtained are compared to those stored in the Pareto archive N_D; the dominated ones are removed and the non dominated ones are added to the set of non-dominated solutions N_D. The Pareto optimal set should be updated, in order to obtain the frontier between dominated solutions and non-dominated solutions. This frontier contains the best tradeoffs between the compared objectives.

4. EXPERIMENTAL ANALYSIS

In order to validate the MOSPACO algorithm, some experiments have been conducted. Our computational model was implemented using C# programming language. The database is build using real geographical coordinates, for a case of intermodal network with 13 cities in Europe (figure 2). The blue lines represent waterways, the black lines the airways, the red lines are the roads and the green lines represent railways.

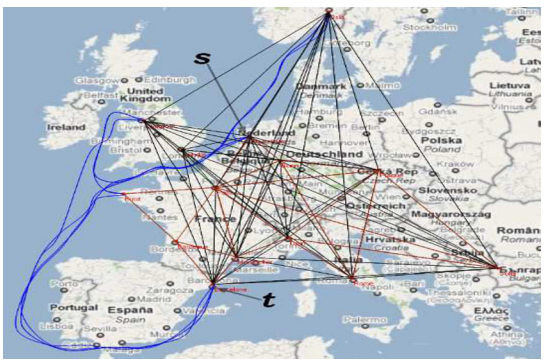


Fig. 2 The intermodal network

The critical MOSPACO parameters α, β, ρ have to be chosen in order to find the best trade-off between the intensification (the reinforcement of the best solutions by increasing the search around promising areas) and the diversification (exploration of the search space in order to avoid the local optima) of the algorithm. The trade-off between diversification and intensification is found by analysing the Pareto front for each couple of criteria. A higher value of the parameter α will increase intensification because pheromone trails will greatly influence the choice of the ants. Conversely, if α is low, the diversification will increase, so α and β have to be choose to find an equilibrium. Diversification can also be obtained by decreasing the evaporation rate ρ; the pheromone evaporates slowly in order to have both diversification and intensification.

Let us apply MOSPACO algorithm to this case study. The value of the parameters are $\alpha=2$, $\beta=1$, $\rho=0.3$, $\chi=0.1$, $N_f=20$ and *iteration=10, Q=50 tonnes*.

The Pareto front is build and some efficient and viable solutions are presented. We consider the transportation of 150 tonnes of good from Roma to Paris. At first glance, it would be easy to make this journey by using an aircraft from Roma to Paris. The transportation cost associated to this alternative is 20161 €, damages 0 tones, travel time 100 minutes, greenhouse gas emitted are 244 tones, energy consumption is 3177 GJ and noise 18 dBA with an accident risk 0.

The obtained solutions are compared according to the scenarios defined above. For the ecological scenario, results are presented in figure 3 and figure 4. In these figures, the red points represent the paths belonging to the Pareto front, and the blue points represent the non-dominated solutions. For noise and greenhouse gas emissions minimization, the non-dominated solution gave a path Roma→ *train*→ Turin→ *train*→ Paris with Noise 58 dBA and greenhouse gas 6.16 tones. For energy consumption and greenhouse gas reduction, one of the paths belonging to the Pareto front is Roma→ *train*→ Turin→ *train*→ Paris with energy consumption 81.82 GJ and greenhouse gas 6.16 tones.

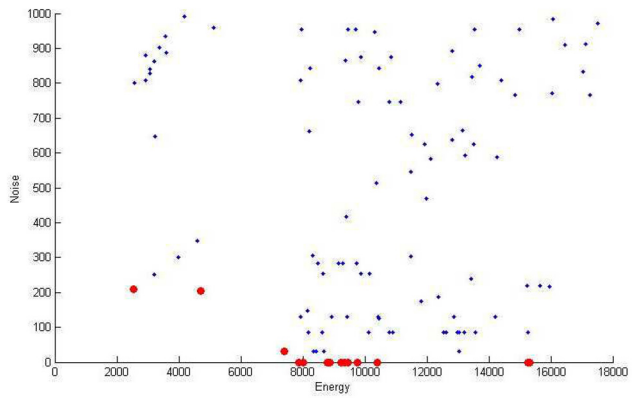


Fig. 3 Pareto front for Energy/noise

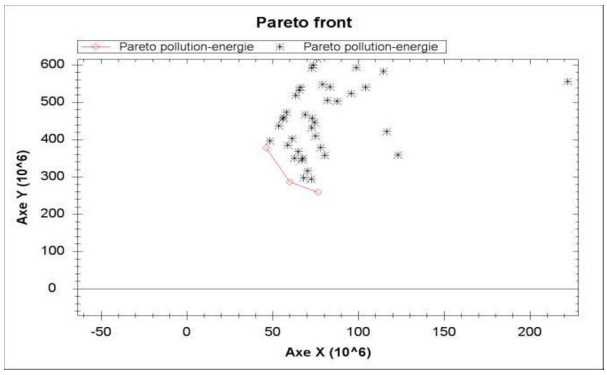


Fig. 4 Pareto front for Pollution/Energy

For the industrial scenario, results are presented in figure 5 and figure 6. For one who wants to simultaneously optimize travel time and cost one of the efficient paths is Roma→ aircraft→ Paris with transportation cost 20162 € and travel time 100 minutes. For damage and travel time reduction, the efficient path is Roma→ aircraft→ Paris damages 0 tone and travel time 100 minutes.

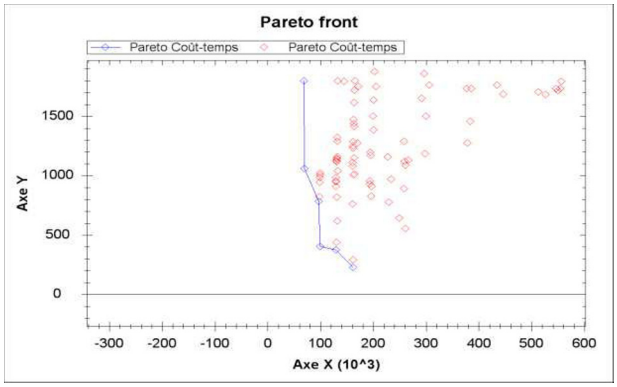


Fig. 5 Pareto front for Cost/Time

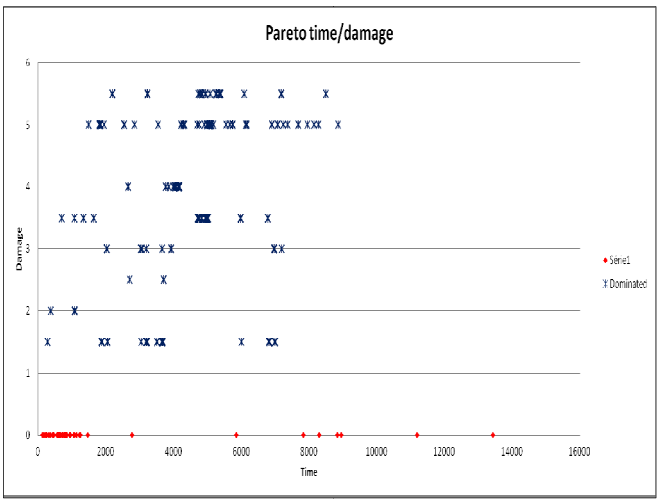


Fig. 6 Pareto front for Time/Damage

By reducing transportation time, the preferred means of transportation in this case would be the aircraft directly from Rome to Paris. However, for greenhouse gas and energy consumption reduction the aircraft is not chosen; the preferred path is by using train through Turin.

A wide choice of solutions belonging to the Pareto front is presented, and then the decision maker can choose among them a path according to his preferences. The performance evaluation of the proposed paths shows that the amount of greenhouse gas and the energy consumption can be reduced considerably. Most the selected paths for environmental impacts reduction are train; these solutions not only help to reduce the impacts. The amount of pollutants reduced compared to the use of aircraft (or another combination of transportation modes such as road) is 237.84 tones, representing a 97% reduction in emissions, the gain in energy consumption is 3088 GJ that to say a 97% reduction in energy consumption. These results are consistent with the actual data that can be found, it shows the truth of this approach and validate the policy choices of the parameters of ACO and the theoretical model underlying.

5. CONCLUSION

The economic climate and the environmental awareness are forcing companies to reconsider the organization of their supply chain, with the aim of favour transportation modes environmentally friendly and insure better coordination of flows to reduce environmental impacts and reduce traffic congestion. In our new approach for multiobjective multimodal ant colony optimization, we have presented a model for route choice in an intermodal network.

The goal of this optimization is to promote more environmentally friendly transport mode and to propose an alternative for building logistic system fulfilling sustainable development requirements. The multiobjective shortest path model presented here help in choosing a 'shortest path' in an intermodal network; 'shortest' here means a path with less environmental, societal and economic impacts. We then provided a real decision support system for policy makers or decision makers who want to reduce their environmental impact when choosing trip choosing or for trip selection. Some of the solutions obtained can bring a gain up to by 237 tone of CO_2 comparing the non-dominated solutions with some dominated solutions. This contribution highlights the fact that a wise choice of transport mode and path contributes to the construction of a sustainable intermodal transportation system. The choice of path is therefore a means of reducing emissions from freight transport.

For future works, we will integrate in our study, time windows for departure time and arrival time choosing in order to avoid delays and for congestion avoidance. Transportation mode practicality, which is the ability of a transportation mode to be practical for the conveyance of Q (tons) or V (m3) of goods over a distance within a time limit set T_{ij}^{m}, must be include as a condition for the choice of transport mode. The number of cities must also be increased in order to find more efficient solutions. It would also be interesting to include the availability of the transportation mode when doing transhipment.

6. REFERENCES

Androutsopoulos, K. N., Zografos, K. G. (2008). Solving the k-shortest path problem with time windows in a time varying network, Operations Research Letters 36, 692-695.

Alaya, I., Solnon, C., Ghedira, K. (2007). Ant Colony Optimization for Multi-Objective Optimization Problems. Proceedings of the 19th IEEE International Conference on Tools with Artificial Intelligence 01, 450-457.

Angus, D., Woodward, C. (2009). Multiple objective ant colony optimization. Swarm Intell 3: 69–85, DOI 10.1007/s11721-008-0022-4.

Barán, B., Schaerer, M. (2003). A multiobjective ant colony system for vehicle routing problem with time windows. In Proceedings of the 21st IASTED international conference on applied informatics. 97–102. Calgary: ACTA Press.

Beamon, B.M. (1999). Designing the Green Supply Chain, Logistics Information Management 12 (4). 332-342.

Beamon, B. M. (2008). Sustainability and the Future of Supply Chain Management. Operations and supply chain management 1 (1). 4-18.

Caramia, M., Guerriero, F (2009). A Heuristic approach to long-haul freight transportation with multiple objective functions. The International Journal of Management Science 37.600-614.

Chapman, C., Ward, S. (1998). Project Risk Management Processes, Techniques and Insights. Wiley, UK.

Doerner, K., Hartl, R.F., Teimann, M. (2003). Are COMPETants more competent for problem solving?-The case of full truckload transportation, Central European Journal of Operations Research 11 (2). 115–141.

Dorigo, M., Maniezzo, V., Colorni, A. (1991). The Ant System: An Autocatalytic Optimizing Process. Technical Report No. 91-016 Revised, Politecnico di Milano, Italy.

Dorigo M., Blum C. (2005). 'Ant colony optimization theory: A survey', Theoretical Computer Science, Vol. 344, pp. 243 – 278.

Dijkstra, E.W. (1959). A note on two problems in connection with graphs. Numerische Mathematik 1. 269–271.

Forkenbrock D. J. (1999). External costs of intercity truck freight transportation. Transportation Research Part A 33. 505-526.

Gambardella, L., Taillard, E., Agazzi, G. (1999). MACS-VRPTW: A multiple ant colony system for vehicle routing problems with time windows, in: D. Corne, M. Dorigo, F. Glover (Eds.), New Ideas in Optimization, McGraw-Hill. 73–76.

Gandibleux, X., Beugnies, F., Randriamasy, S. (2006). Martins' algorithm revisited for multi-objective shortest path problems with a MaxMin cost function. A Quarterly Journal of Operations Research 4 (1), 47-59.

García-Martínez, C., Cordón, O., Herrera, F. (2007). A taxonomy and an empirical analysis of multiple objective ant colony optimization algorithms for the bi-criteria TSP. European Journal of Operational Research 180. 116–148.

Ghoseiri, K., Nadjari, B. (2010). An ant colony optimization algorithm for the bi-objective shortest path problem. Applied Soft Computing 10. 1237–1246.

Gravel, M., Price, W. L., Gagné, C. (2002). Scheduling continuous casting of aluminum using a multiple objective ant colony optimization metaheuristic. European Journal of Operational Research 143 (1). 218–229.

Guerriero F., Musmanno R. (2001). 'Label Correcting Methods to Solve Multicriteria Shortest Path Problems', Journal of Optimization Theory and Applications: Vol.111, No. 3, pp.589-613.

Janic, M. (2007). Modelling the full costs of an intermodal and road freight transport network. Transportation Research Part D 12. 33-44.

Klassen R, Johnson P.F. (2004). *The green supply chain. In Westbrook, R. & New, S. (Eds.)*. Understanding Supply Chains - Concepts, Critiques and Futures. p. 229 -251.

Kostreva, M.M., Wiecek, M.M. (1993). Time dependency in multiple objective dynamic programming. Journal of Mathematical Analysis and Applications 173 (1). 289–307.

Mariano, C.E., Morales, E. (1999). MOAQ: An Ant-Q algorithm for multiple objective optimization problems, in: W. Banzhaf, J. Daida, A.E. Eiben, M.H. Garzon, V. Hnavar, M. Jakiela, R.E. Smith (Eds.), Proc. of the Genetic and Evolutionary Computing Conference (GECCO 99), San Francisco, California, USA, July. 894–901.

Martins, E.Q.V. (1984). on a multicriteria shortest path algorithm, European Journal of Research 16 (2). 236-245.

Martins, E.Q.V., Pascoal, M.M.B., Dos Santos, J.L.E. (1998). The *K* Shortest path Problem.

Modesti P., Sciomachen A. (1998). 'A utility measure for finding multiobjective shortest paths in urban multimodal transportation networks', European Journal of Operational Research, Vol.111, pp.495-508.

Pangilinan, J.M. A., Janssens, G. K. (2007). Evolutionary Algorithms for the Multiobjective Shortest Path Problem. International Journal of Computer and Information Science and Engineering 1 (1).

Piecyk, M.I., McKinnon, A.C. (2010). Forecasting the carbon footprint of road freight transport in 2020. International Journal of Production Economics 128 (1). 31-42.

Qu, L., Chen, Y. (2008). A Hybrid MCDM Method for Route Selection of Multimodal Transportation Network. Part I, LNCS 5263. 374 – 383.

Quariguasi Frota Neto, J., Walther, G., Bloemhof, J., van Nunen, J.A.E.E. and Spengler, T. (2009). A methodology for assessing eco-efficiency in logistics networks. European Journal of Operational Research 193 (3). 670-682.

Rondinelli D, Berry M. (2000). Multimodal Transportation, Logistics, and the Environment: Managing Interactions in a Global Economy. European Management Journal 18. 398–410.

Saaty, T. (1990). How to make a decision: The Analytic Hierarchy Process. European Journal of Operational Management 48. 9-26.

Sawadogo, M., Anciaux, D. (2009). Intermodal transportation within the green supply chain: An approach based on the ELECTRE method. Computers & Industrial Engineering (CIE), International Conference on Industrial Engineering. 839 – 844.

Sawadogo, M., Anciaux, D. (2010). Modèle de plus court chemin multiobjectif pour le transport intermodal au sein de la chaîne logistique verte. Conférence Internationale de MOdélisation et SIMulation - MOSIM'10 - 10 au 12 mai 2010 - Hammamet - Tunisie.

Skriver A.J.V., Andersen, K.A. (2000). A label correcting approach for solving bicriterion shortest-path problems. Computers & Operations Research 27. 507-524.

Srivastava, S. K. (2007). Green supply-chain management: A state-of-the-art literature review. International Journal of Management Reviews 9 (1). 53–80.